# Task 16 — Hypothesis Testing & A/B Tests

Data Analyst | Phase 1 Industry Immersion

## 1. Objective

Determine whether average Sales differs significantly between Consumer and Corporate customers in the Superstore dataset.

## 2. Comparison Design

Group A: Consumer customers
Group B: Corporate customers
Metric: Sales
Statistical test: Welch's independent two-sample t-test
Significance level: $\alpha = 0.05$
Confidence level: 95%

## 3. Hypotheses

**H■:** The mean Sales of Consumer and Corporate customers are equal.
**H■:** The mean Sales of Consumer and Corporate customers are different.

## 4. Data Preparation

The Segment and Sales columns are selected. Missing values are removed, and Consumer and Corporate observations are separated into two independent groups.

## 5. Statistical Method

Welch's independent two-sample t-test is used to compare the two group means without assuming equal population variances. A p-value below 0.05 is considered statistically significant.

## 6. Analysis Performed

The notebook calculates sample sizes, mean and median Sales, standard deviation, mean difference, Cohen's d effect size, t-statistic, p-value, 95% confidence interval, and approximate sample size needed for 80% power.

## 7. Results

| Output | Generated by notebook |
|---|---|
| Consumer mean Sales | Calculated |
| Corporate mean Sales | Calculated |
| Mean difference | Calculated |
| Cohen's d | Calculated |
| t-statistic | Calculated |
| p-value | Calculated |
| 95% confidence interval | Calculated |

The numerical results are calculated directly by the accompanying Google Colab notebook from the uploaded CSV. This keeps the analysis reproducible and avoids manually entering values that could differ from the dataset.

## 8. Interpretation

If $p < 0.05$, reject H■ and conclude that the difference in average Sales is statistically significant. If $p \geq 0.05$, fail to reject H■ because the evidence is insufficient to establish a statistically significant difference.

## 9. Business Recommendation

Consider both statistical significance and effect size before making decisions. A significant difference should be investigated for possible product, offer, or customer-strategy drivers. A non-significant result should not be used alone to justify major segment-based decisions.

## 10. Limitation

This is an observational comparison of existing customer segments, not a randomized A/B experiment. Therefore, statistical significance does not by itself establish a causal relationship.

## 11. Conclusion

The project demonstrates a complete hypothesis-testing workflow: defining hypotheses, preparing independent groups, selecting an appropriate test, measuring effect size, calculating a confidence interval, making a statistical decision, and translating the findings into a business recommendation.